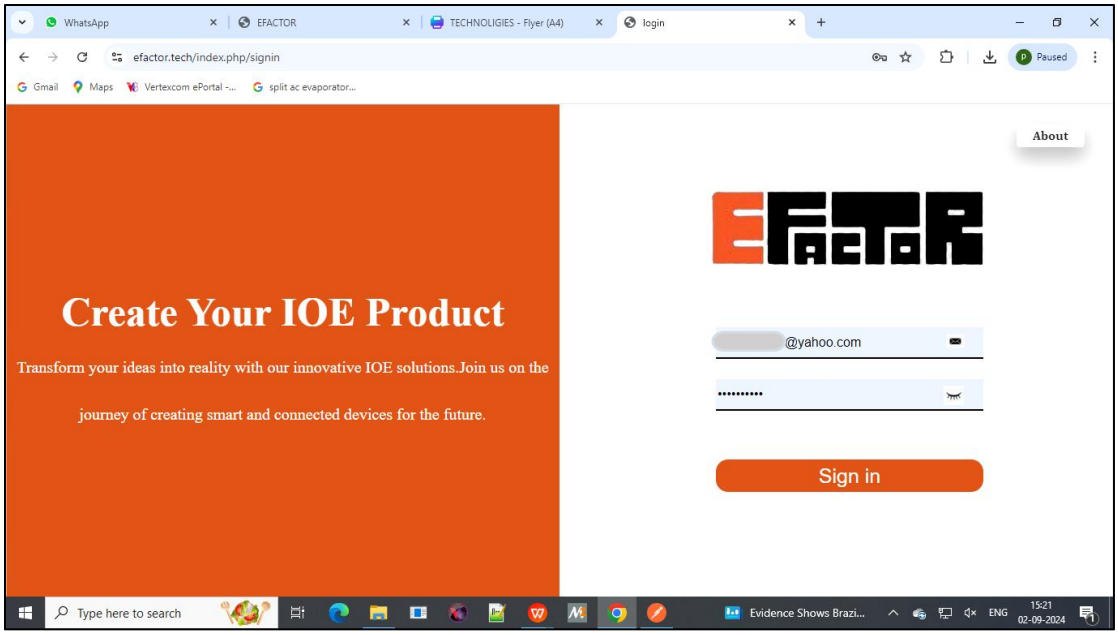
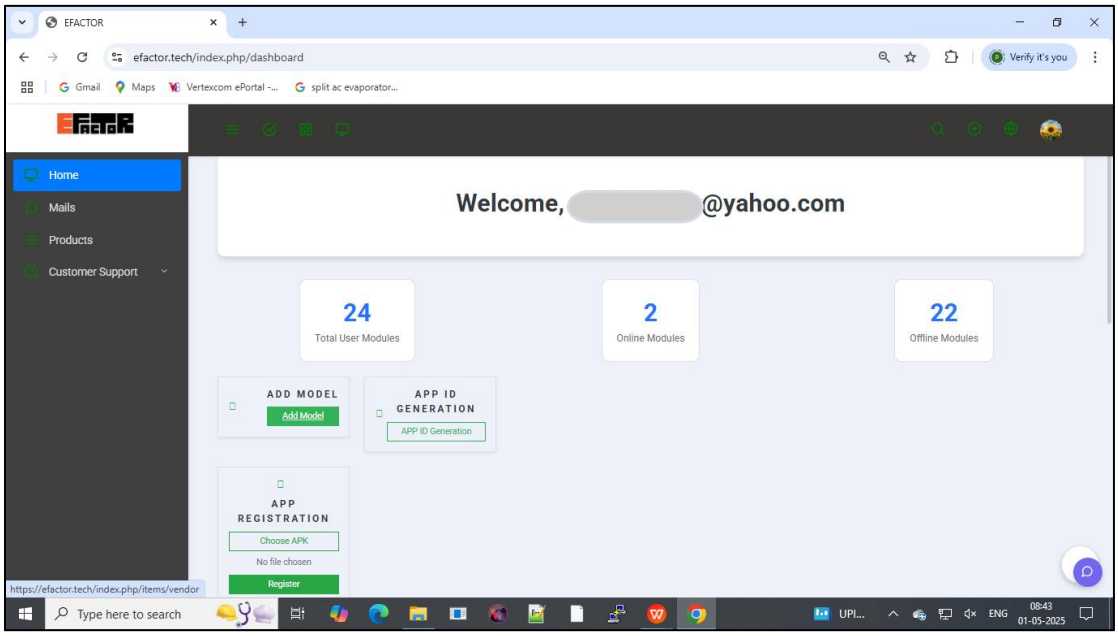
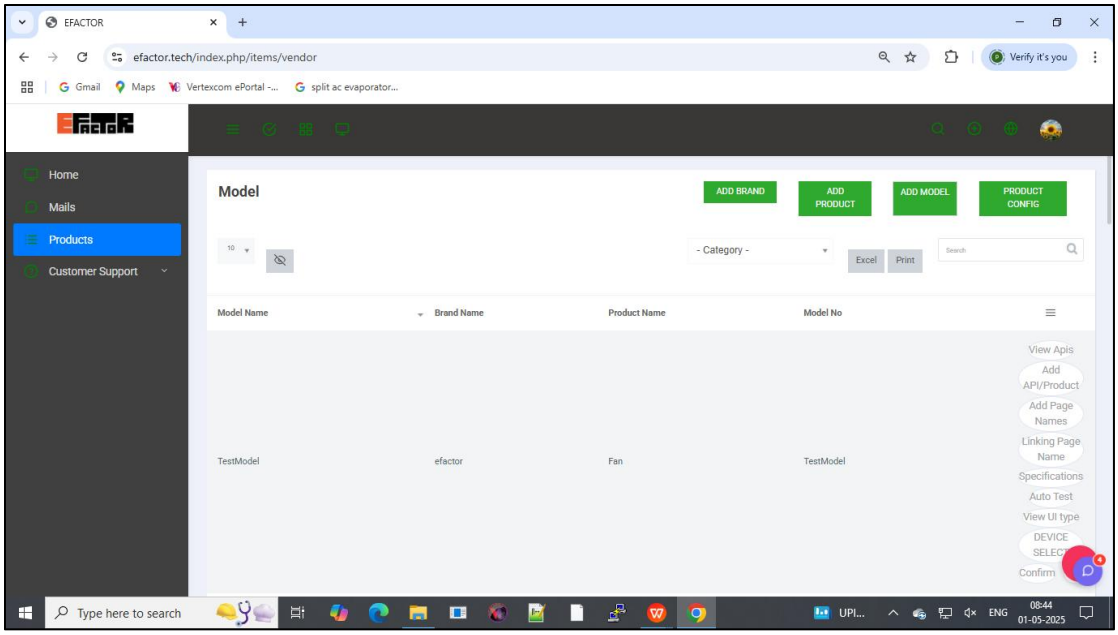


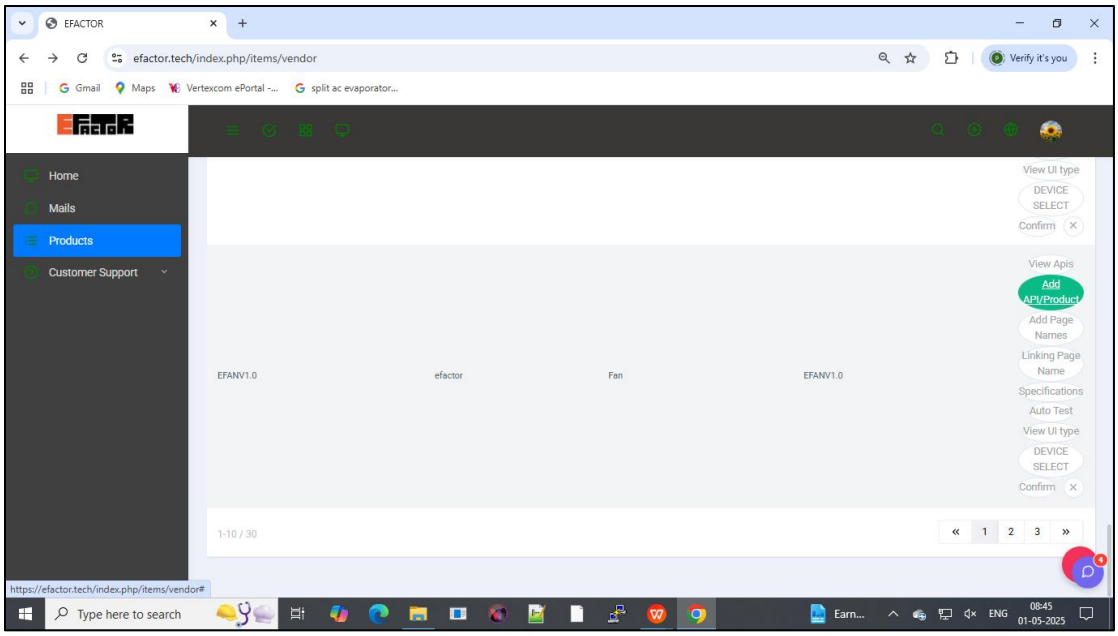
- This is “EFACTOR Studio” and we need to “Sign in” in this page by entering “Email Id” and “Password”.



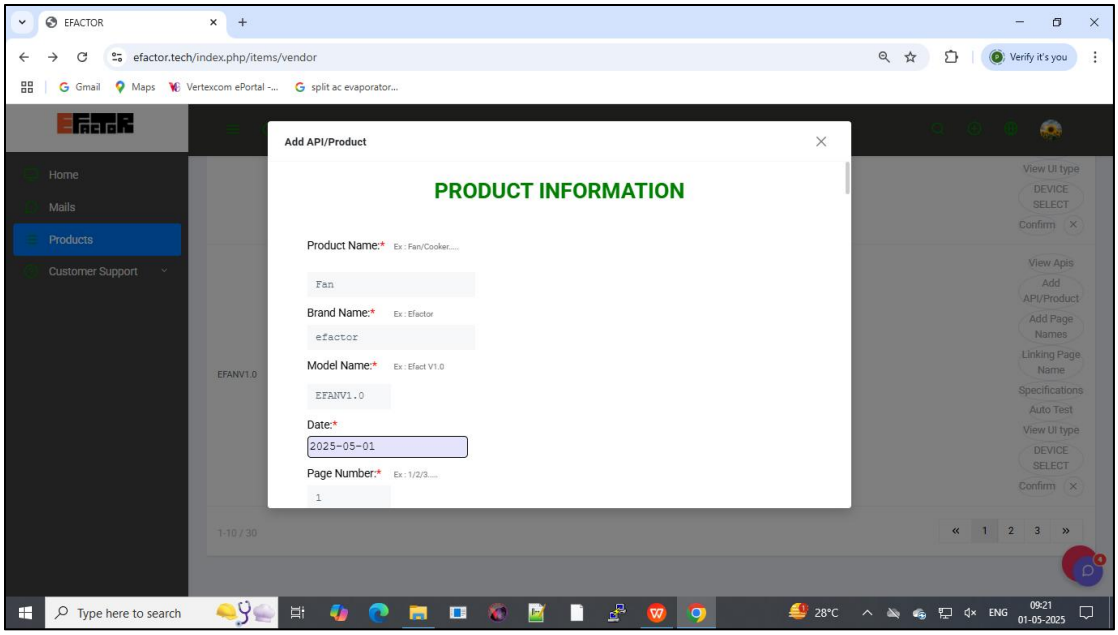
- This is the “Home Page” of the EFACTOR Studio.
- Here we can see “Total User Modules , Online Modules, Offline Modules, Add Brand, Add Product and Add Model”.
- In Left side we can see “Home Page, Mails, Products and Customer Support”.
- Click on “Add Model” to Add the “Brand, Model, Product and Product Config”.



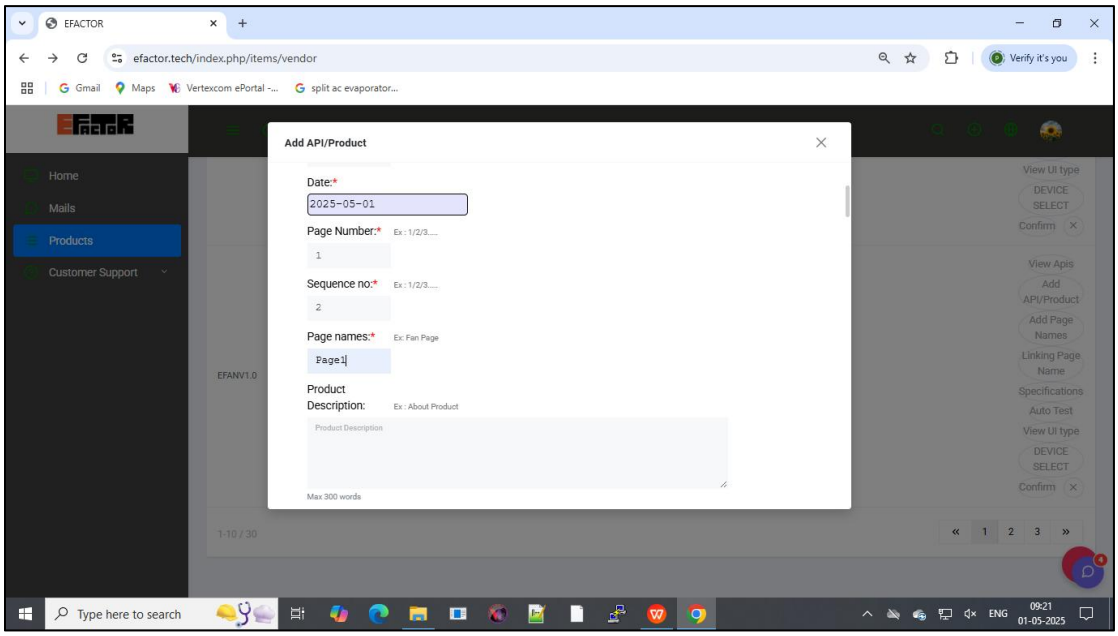
- This is “Add Model” Page.
- This page allows you to add a “brand, product, model, and product config”.
- If these elements have already been added, and you wish to add a “UI”.
- Simply select the appropriate “model name” and “product name” where the UI needs to be added.



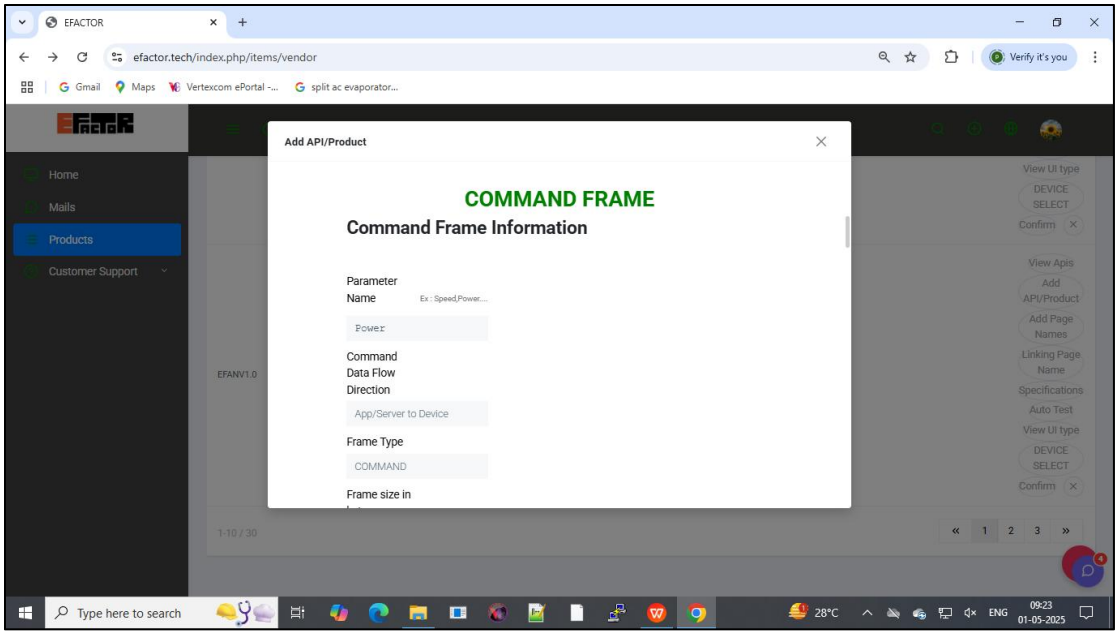
- I have selected the “EFANV1.0” model and the “Fan” product to add a UI to this configuration.
- To proceed, click on “Add API/Product” to integrate the “UI”.



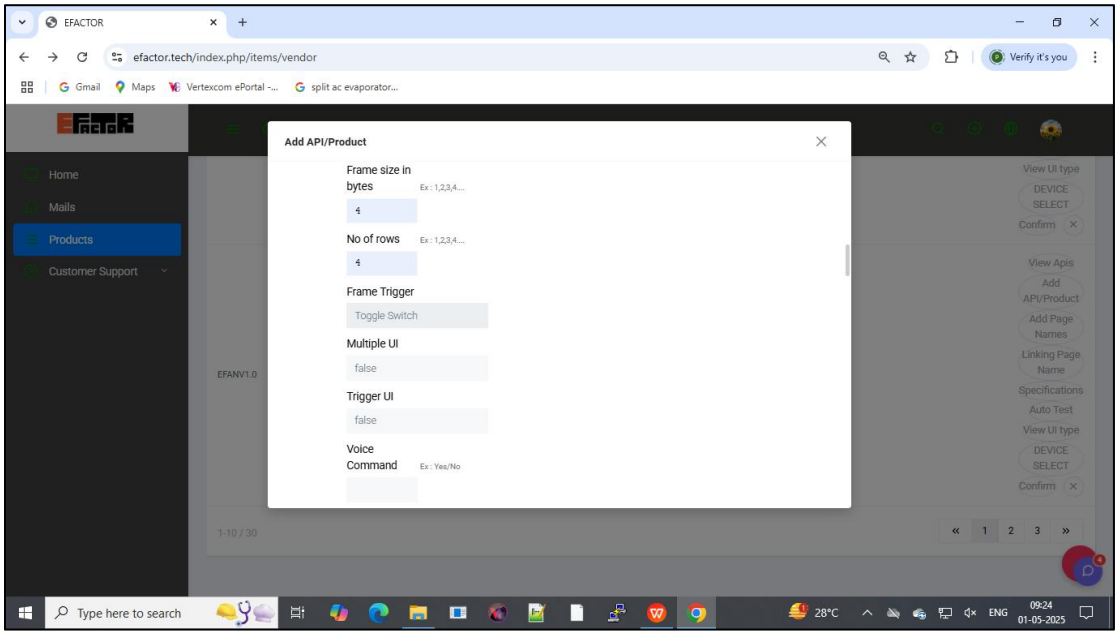
- After clicking on "Add API/Product", a popup window will appear.
- In this popup, you are required to enter the necessary UI details.
- The fields for “Product Name, Brand Name, Model Name, and Date” will be automatically populated.



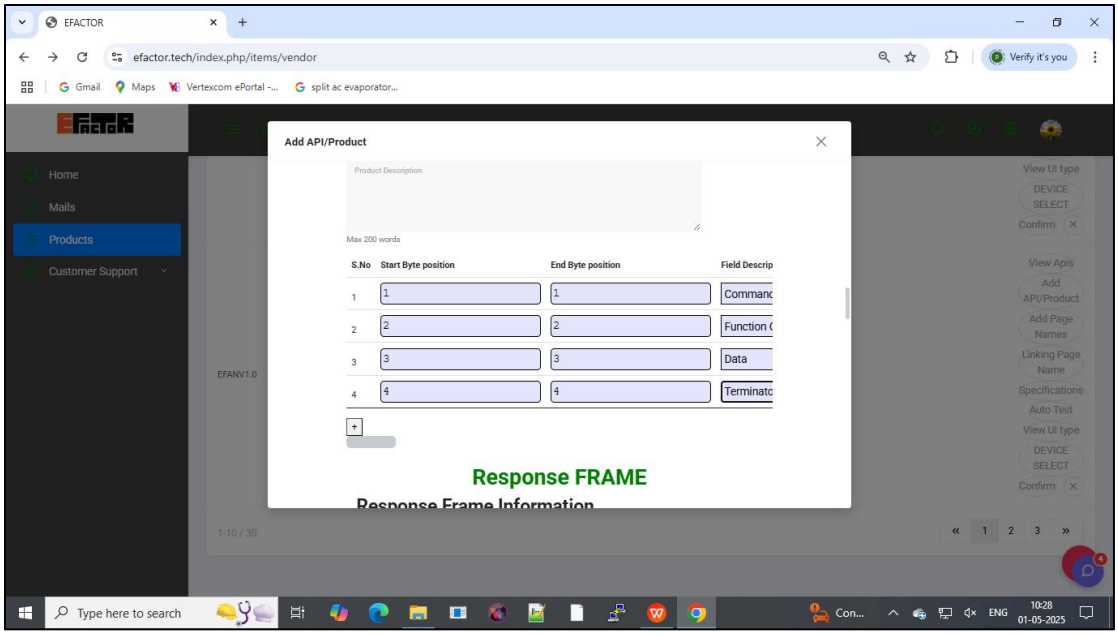
- Scroll down to enter the “Page Number, Sequence Number, and Page Name”.
- Page Number: Specify the page where this UI should appear.
- Sequence Number: Define the row position for this UI.
- Page Name: Assign a custom name for the page as desired.



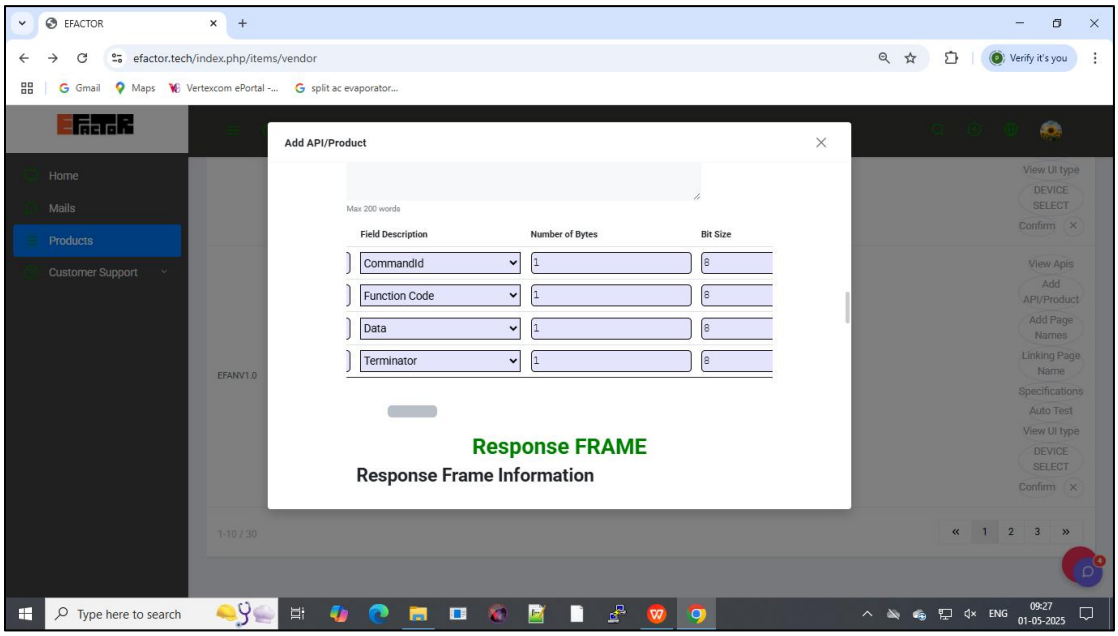
- After entering the values, scroll down to the “Command Frame” section.
- Enter the required “Parameter Names” (e.g., Power, Name, Voltage, Current, etc.).
- For example, "Power" is used as a sample parameter name.



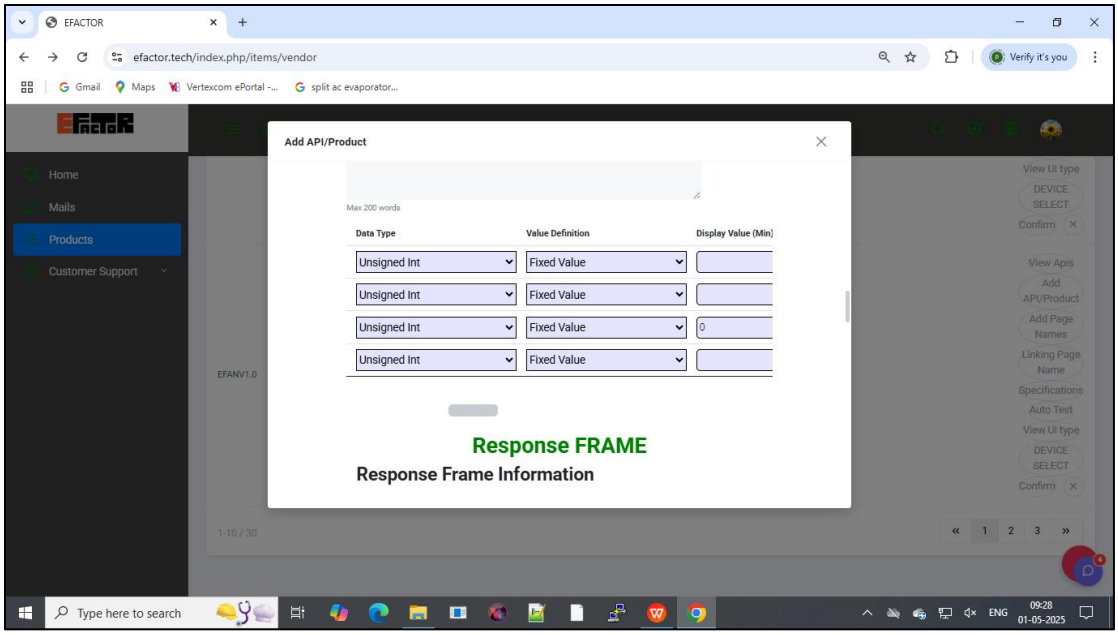
- Enter “Frame Size, No. of Rows, and Frame Trigger” after the Parameter Name.
- Example: Power → 4 bytes, 4 rows.
- In Frame Trigger, select “Toggle Switch” (ON/OFF) for Power.
- Set "Multiple UI" to true if the parameter has multiple UIs.
- If it's a Trigger UI, set "Trigger UI" to true.



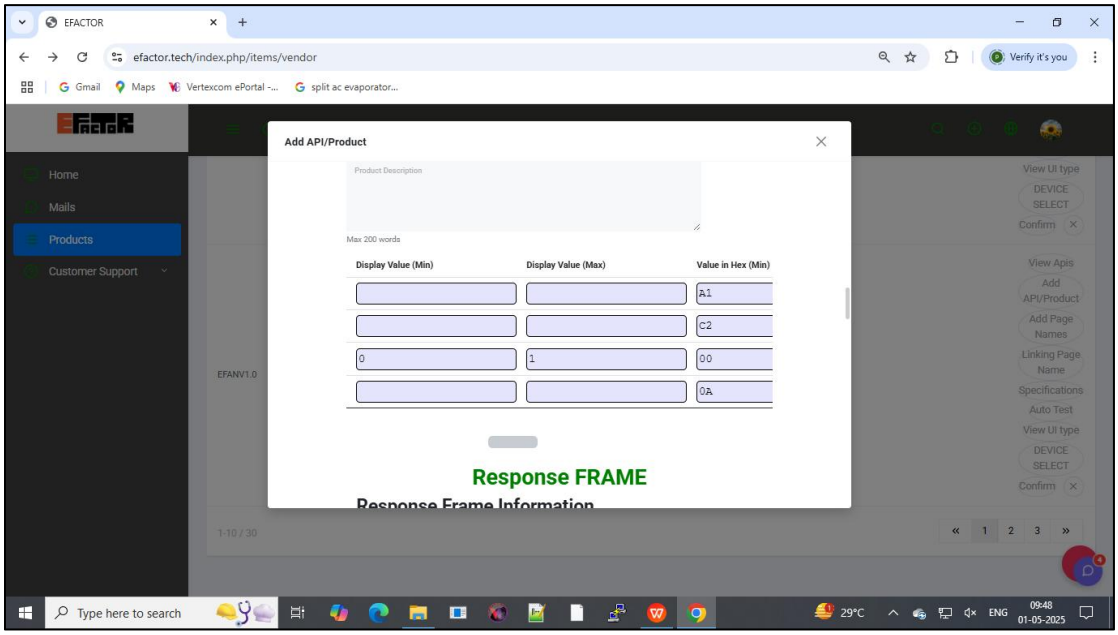
- Input Byte Details: Scroll down to locate the 4-row entry section for byte positions.
- Define Byte Range: Each row corresponds to a data field (e.g., Power – 4 bytes).Specify start & end byte positions accordingly.
- Single-Byte Entry: For fields with only 1 byte, enter the same value for start and end(e.g.,Serial No. 1: 1–1, Serial No. 2: 2–2).



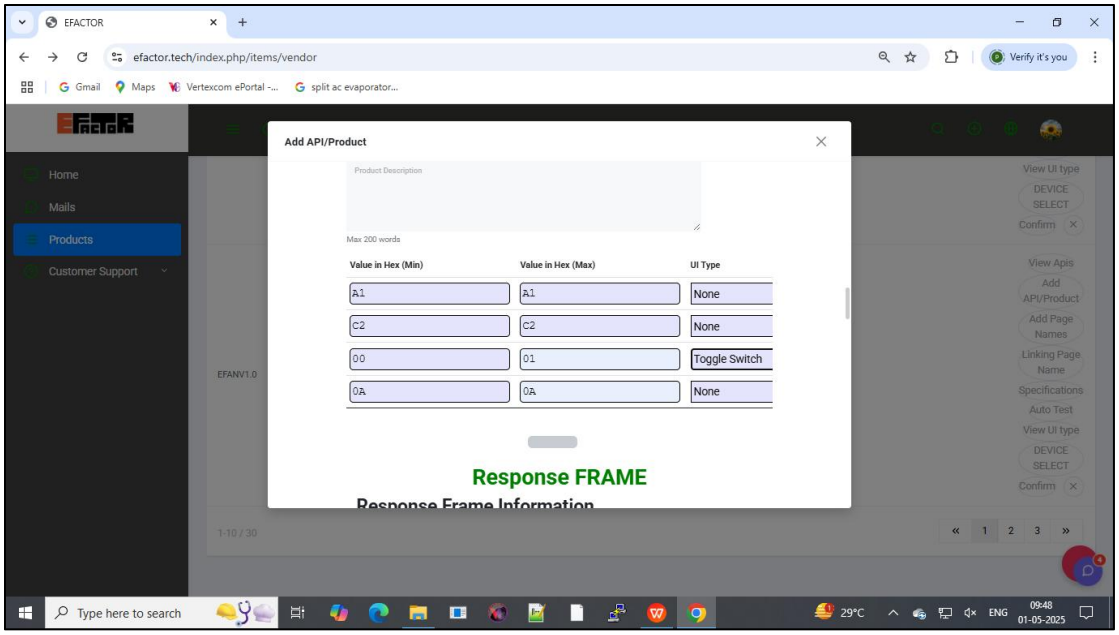
- Field Descriptions: Scroll horizontally to enter field names—e.g., 1st: Command ID, 2nd: Function Code, 3rd: Data, 4th: Terminator.
- Number of Bytes: Based on start-end byte ranges (e.g., 1–1, 2–2), input byte count as 1 for each row.
- Bit Size: Note that 1 byte = 8 bits for each field.



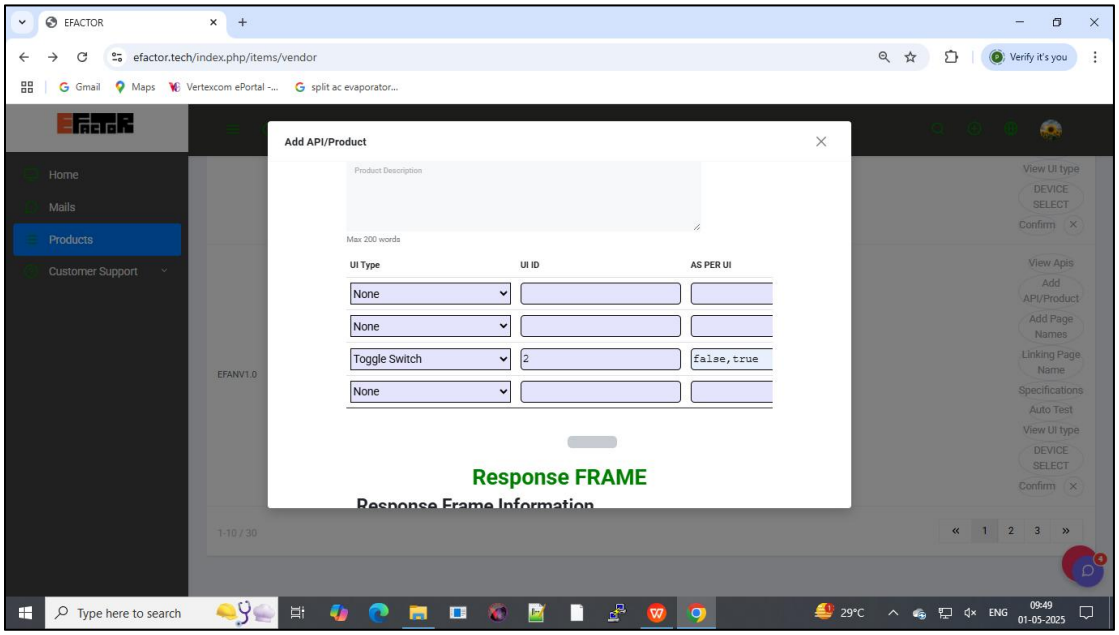
- Data Type Entry: Scroll horizontally to enter Data Type and Value Definition for each field.
- Data Type: Use Unsigned Int for Command ID, Function Code, and Terminator. Data type varies by parameter—Power uses Unsigned Int.
- Value Definition: Set Command ID, Function Code, and Terminator always as Fixed Values. Data may be a Range (e.g., 1–100) or Fixed (e.g., 0 = OFF, 1 = ON). Power uses fixed ON/OFF values.



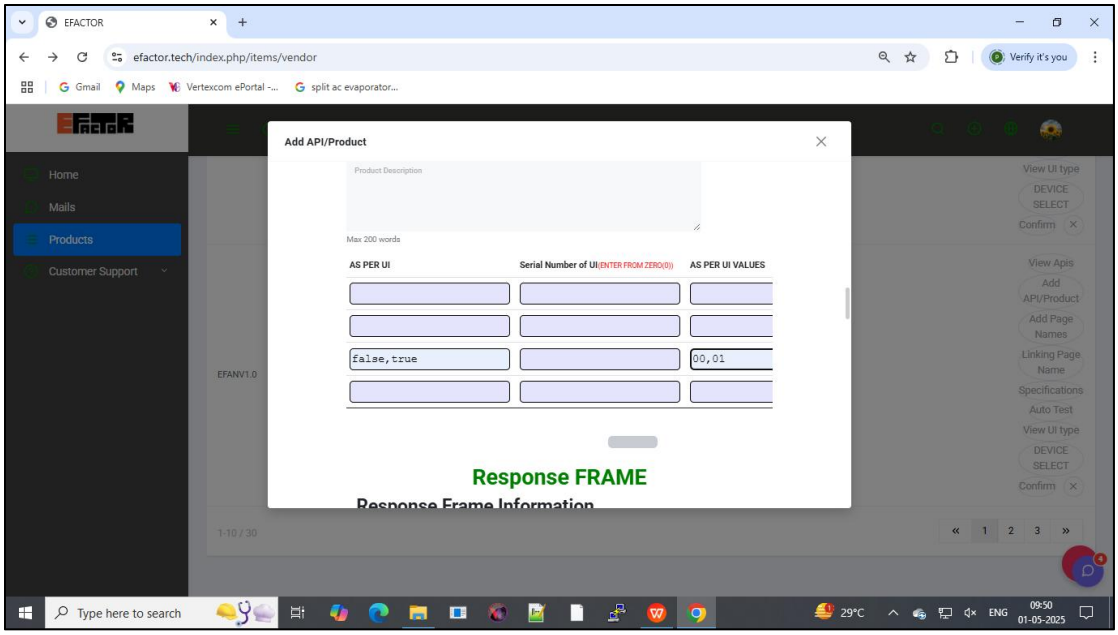
- Display Value (Min–Max): Specify min–max values (e.g., 0–100) to set visible limits for the parameter.



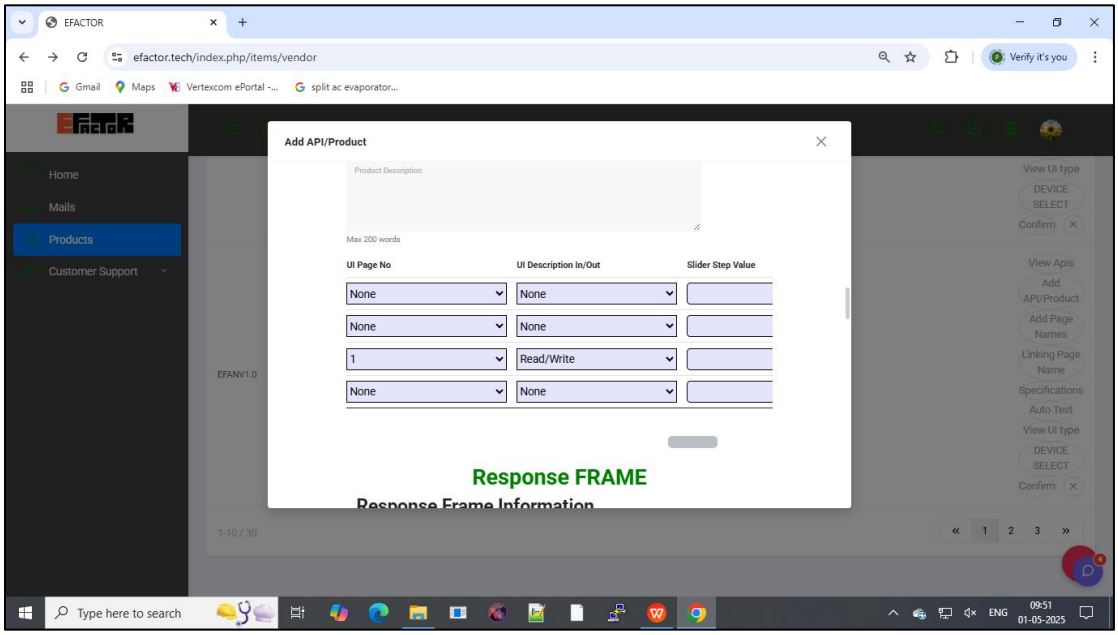
- Value in Hex(Min–Max): For Command ID, Function Code, and Terminator, use fixed hex values. For Data, enter min–max hex range based on the parameter (e.g., Power: 00/01 for OFF/ON).



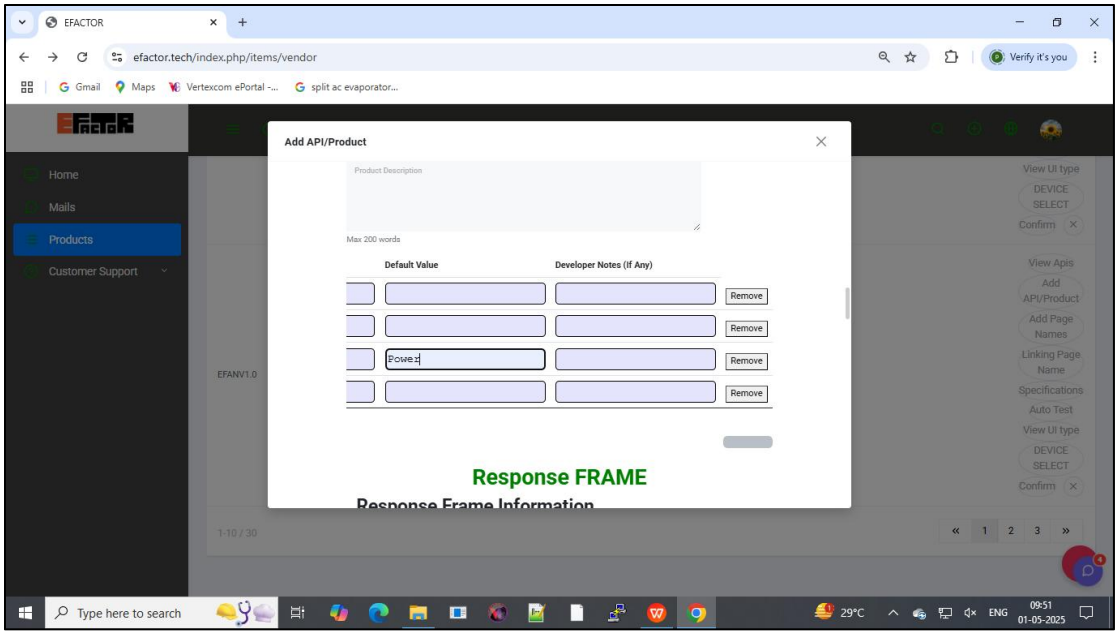
- UI Type: Apply UI only for Data (others = None); use “Toggle Switch” for Power (OFF/ON).
- UI ID: Set UI ID only for Data by specifying the app screen row number; leave others blank.



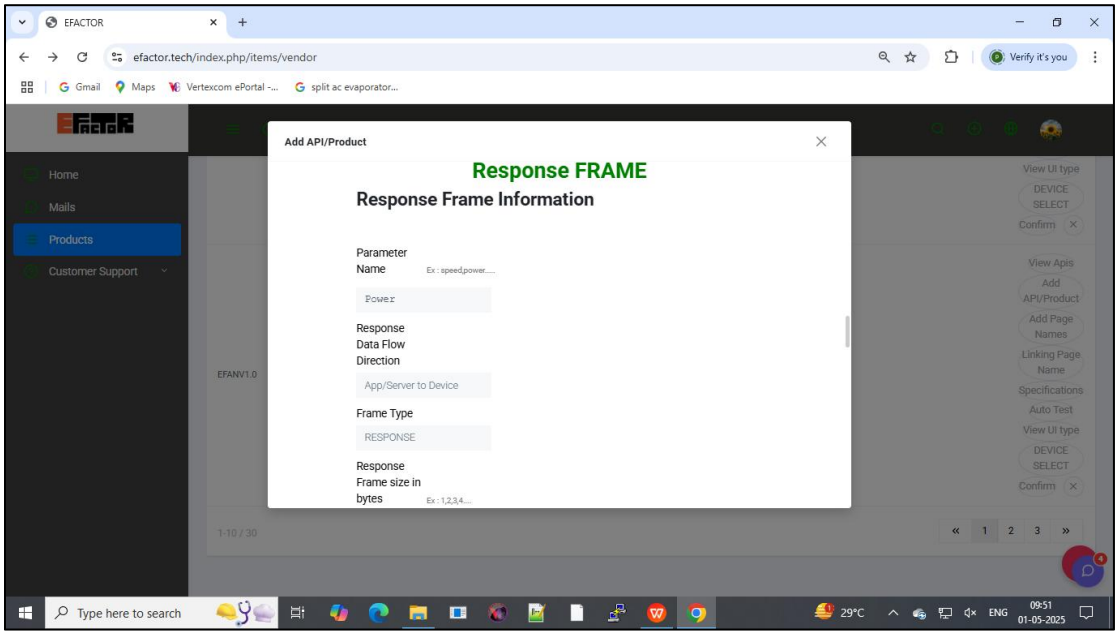
- As Per UI: Enter display names/labels for UI elements (e.g., Toggle Switch, Dropdown) shown on screen like False,True.
- As Per UI Values: Map each display label to its exact Hex value like 00,01 (e.g., 00 = False, 01 = True).



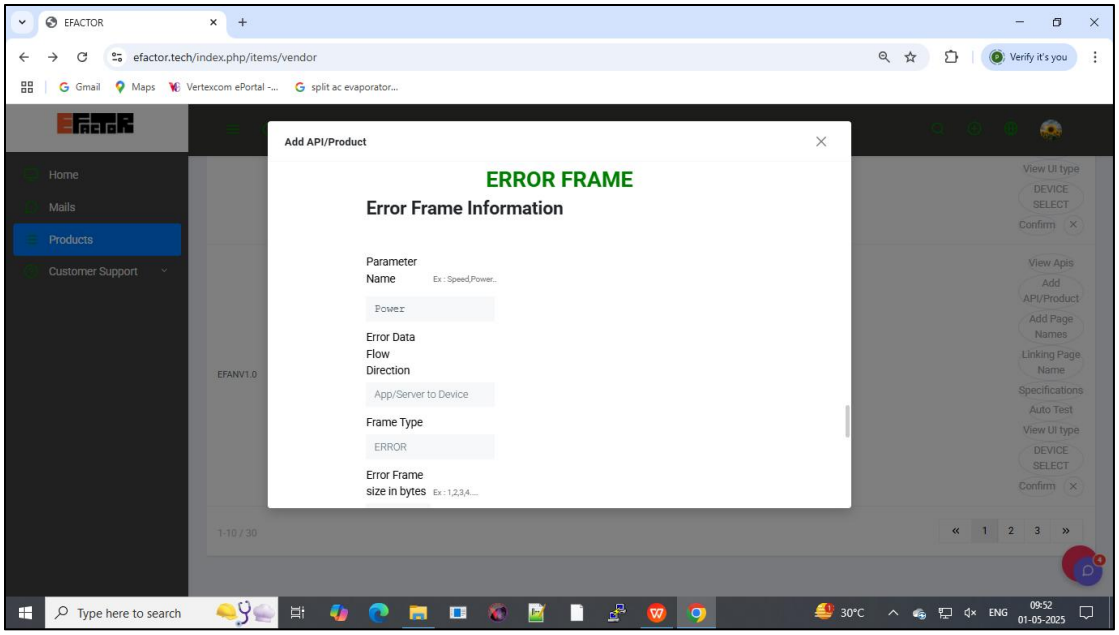
- UI Page No: Specify the page number where the UI should appear (e.g., 1 = Page 1).
- UI Description (In/Out): Enter only for Data—choose Read, Write, or Read/Write. For Power, use Read/Write to allow ON/OFF control and status display.



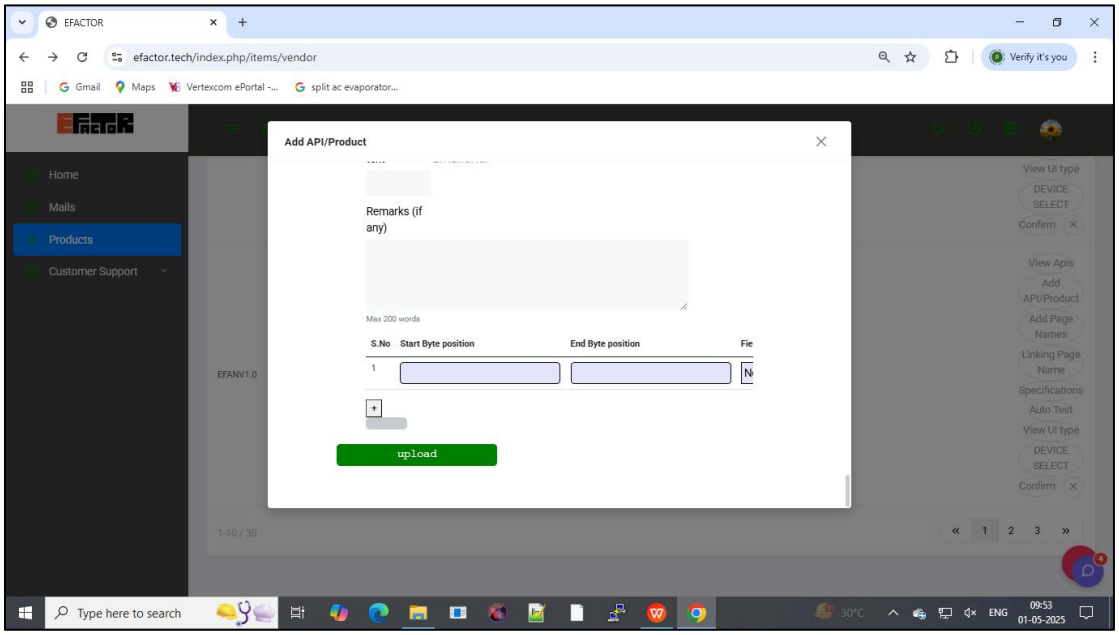
- Default Value: Enter the UI name to be displayed on the app screen; typically use the Parameter Name as the default.



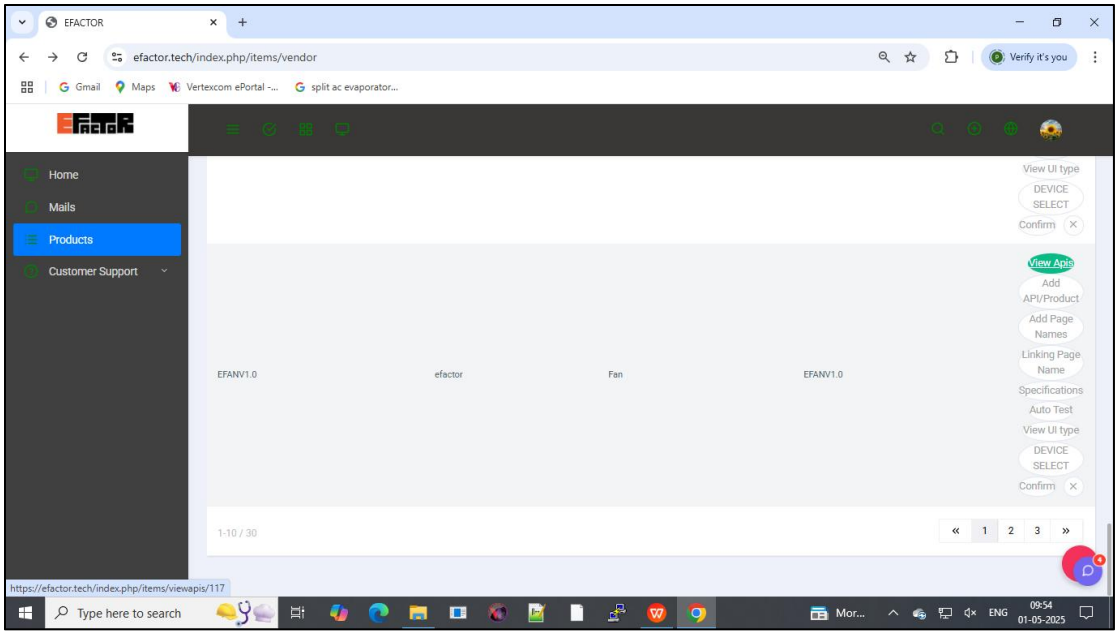
- Response Frame: Scroll down and add the same parameter name used in the Command Frame. No additional data needed for Power—proceed directly to the Error Frame after entering the name.



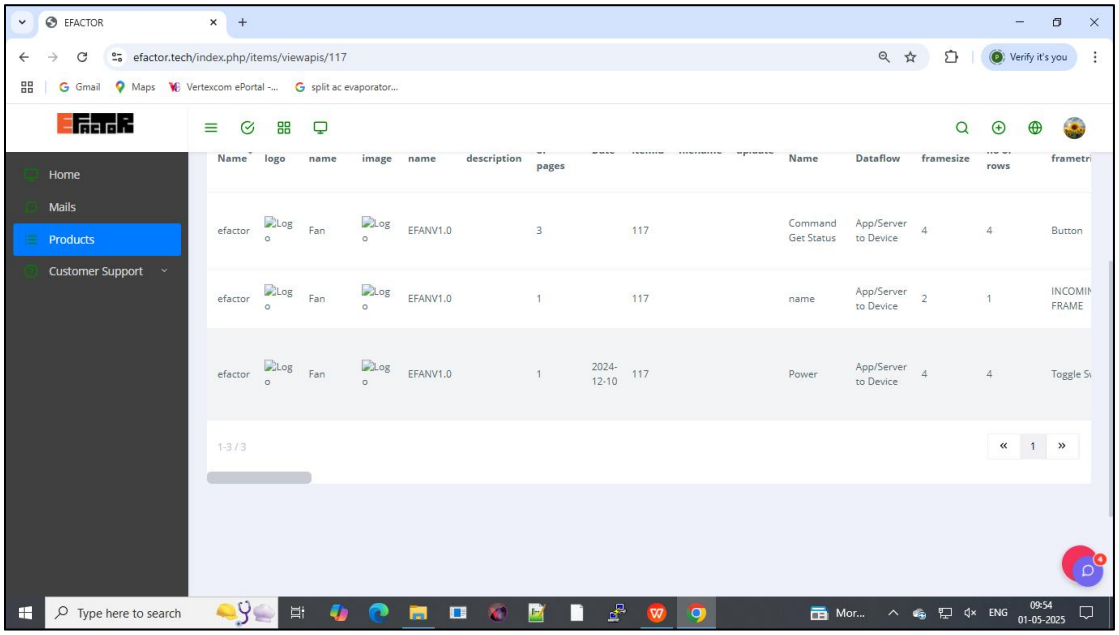
- Error Frame: Scroll down and add the same parameter name used in the Command Frame. No additional data needed for Power—just proceed.



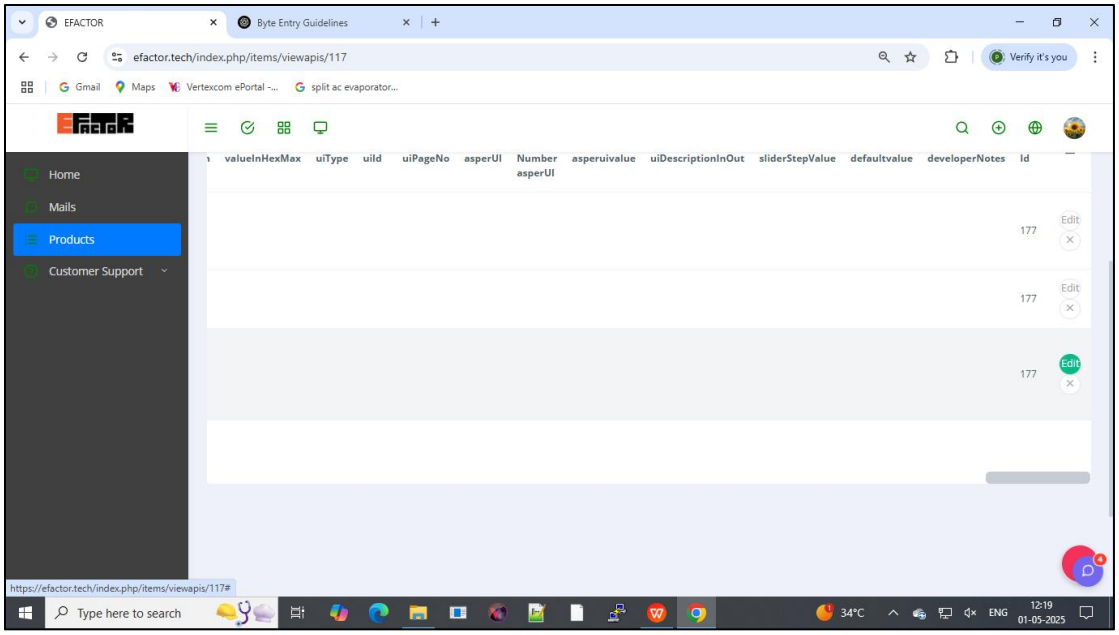
- Upload: Click the Upload button directly; the popup will close automatically after clicking.



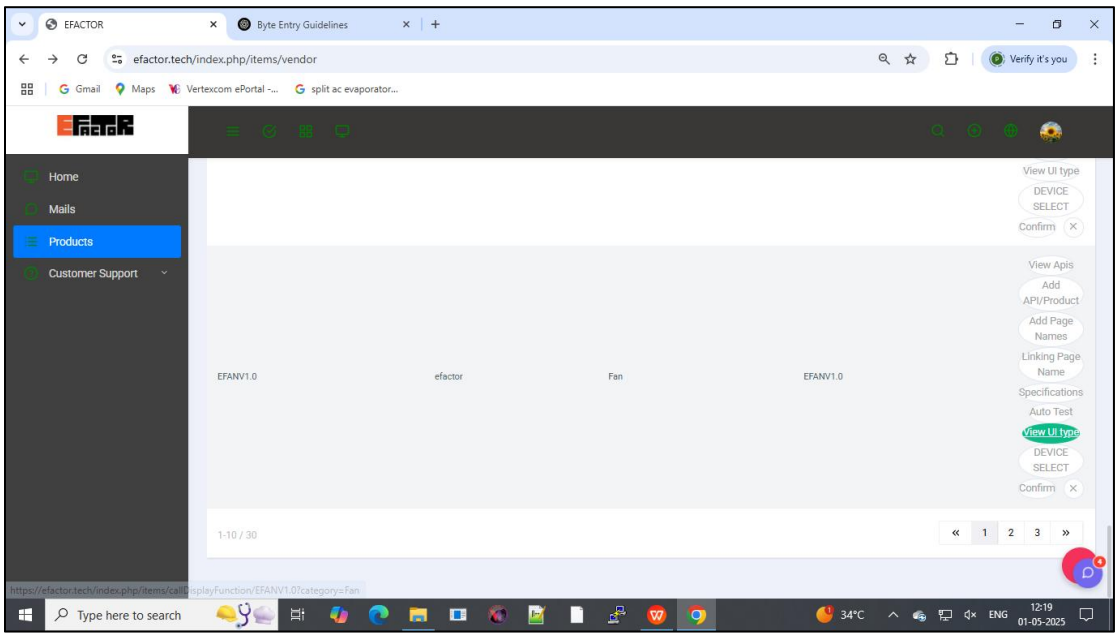
- View Api's: To edit previously entered UI details, simply click on View APIs.



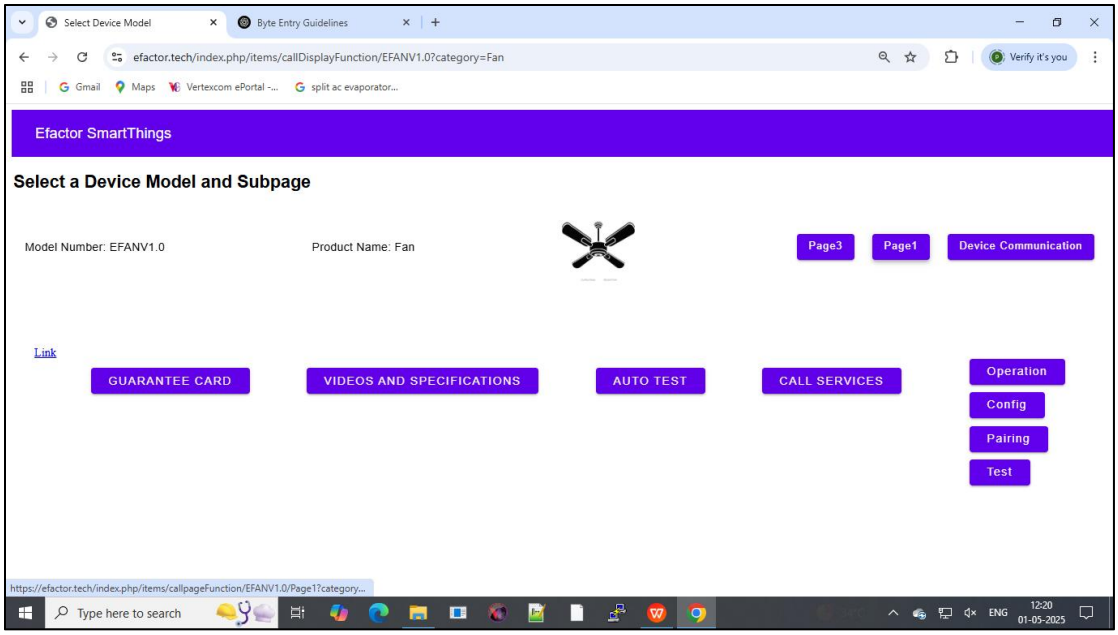
- View APIs Page: After clicking View APIs, you'll see a list of UIs (e.g., Command Get Status, Name, Power).
- Scroll horizontally to the end of the row to access the Edit option for the desired UI.



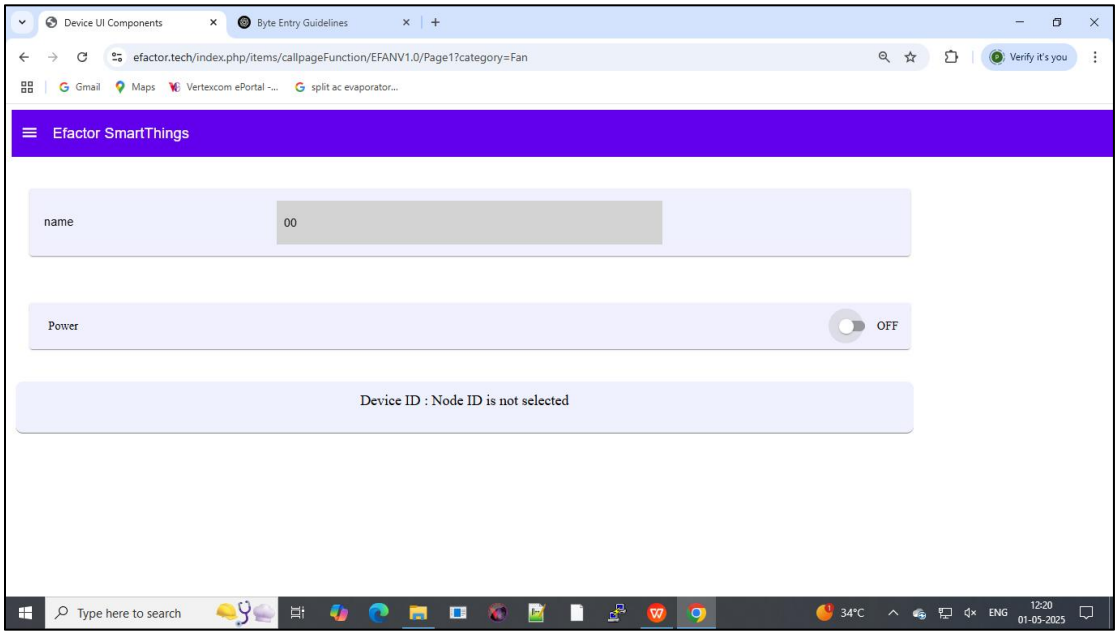
- Edit: After scrolling, click the Edit option to modify the desired UI settings as needed.



- View UI Type: Click View UI Type to verify if the UI has been added correctly to the intended page.



- Select Page: On the View UI Type page, click the page number where the UI was added (e.g., Page 1 for Power).
- Verify UI: Check if the selected UI (e.g., Power) is correctly displayed on that page.



- Page1: When you click on Page 1, you'll see the "Power" UI and its associated "Toggle Switch".
- This confirms that the UI has been successfully added to the selected page.